COMP0082 Coursework Report — Protein Enzyme Classification

March 2026

1 Introduction

Enzymes are biological catalysts that underpin virtually every metabolic process in living organisms. The Enzyme Commission (EC) numbering system classifies enzymes into six top-level classes— Oxidoreductases (EC 1), Transferases (EC 2), Hydrolases (EC 3), Lyases (EC 4), Isomerases (EC 5), and Ligases (EC 6)—according to the reaction they catalyse. Computational prediction of enzyme class from amino acid sequence alone is of substantial practical importance: it accelerates the functional annotation of the millions of uncharacterised proteins generated by high-throughput sequencing, and it supports drug target identification, metabolic engineering, and synthetic biology.

This project addresses the problem as a seven-class supervised classification task, where class 0 represents *Not an enzyme* and classes 1–6 correspond to the six EC classes. The dataset comprises 39,764 non-homologous protein sequences with a heavily skewed distribution: class 0 constitutes 81.5% of the data, while the smallest class (Ligase, EC 6) accounts for only 0.7%. This extreme class imbalance—spanning two orders of magnitude between the largest and smallest classes—is the central challenge of the project and informs every design decision, from loss weighting and oversampling strategies to the choice of evaluation metrics.

The approach progresses systematically from handcrafted sequence-based features through protein language model (PLM) embeddings to an ensemble that combines the complementary strengths of gradient-boosted trees and end-to-end fine-tuned transformers. All experiments use stratified five-fold cross-validation with strict data-leakage prevention, and all results are reported with standard deviations across folds to quantify variability.

Table 1: Dataset class distribution.

Class	Label	Count
0	Not an enzyme	32410
1	Oxidoreductase	1184
2	Transferase	2769
3	Hydrolase	2108
4	Lyase	600
5	Isomerase	411
6	Ligase	282

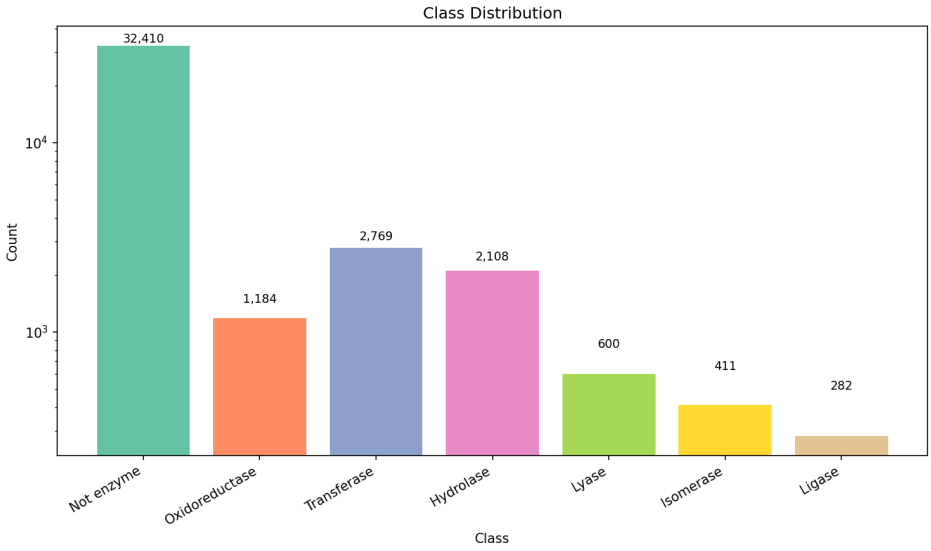


Figure 1: Observed class imbalance in the training dataset.

2 Methods

2.1 Feature Engineering

Amino acid composition (20-d). The normalised frequency of each canonical amino acid, capturing global residue distribution. This is the simplest possible sequence representation and serves as a well-established baseline in protein classification literature.

Dipeptide frequencies (400-d). Normalised counts of all ordered amino acid pairs, encoding local sequential context that single-residue composition cannot capture. Combined with a sequence-length scalar, this yields a 421-dimensional handcrafted composition vector.

Physicochemical properties (8-d). Molecular weight, isoelectric point, aromaticity, GRAVY hydropathicity, charge at pH 7.0, and Chou–Fasman secondary-structure propensity fractions (helix, turn, sheet), computed using BioPython ProtParam. Unlike composition features, these capture higher-level biophysical properties that may distinguish enzyme families with different catalytic environments.

Protein language model embeddings. ESM-2 (Lin et al., 2023), a transformer-based PLM pre-trained on millions of protein sequences via masked language modelling, was used as a feature extractor. Two model sizes were evaluated: `esm2_t6_8M_UR50D` (320-d) and `esm2_t33_650M_UR50D` (1,280-d). Final-layer hidden states were mean-pooled across non-special tokens. The key advantage of PLM embeddings over handcrafted features is that they encode evolutionary and structural information learned from millions of diverse protein families, capturing patterns that are impossible to design manually.

2.2 Experimental Design

All experiments used stratified 5-fold cross-validation (seed 42). Leakage prevention rules were enforced:

1. Fold indices generated before feature normalisation.

2. `StandardScaler.fit()` on training fold only.
3. SMOTE on training fold only.
4. Features derived only from amino acid sequence.
5. Precomputed ESM embeddings treated as stateless features; normalisation still fold-local.

Metrics: Accuracy, Macro F1, Balanced Accuracy, MCC. Macro F1 was the primary model-selection criterion.

2.3 Models

Baselines: Logistic Regression (`class_weight='balanced'`, multinomial, L-BFGS solver, `max_iter=1000`) and Random Forest (`n_estimators=300`, `class_weight='balanced'`) were trained on the 429-dimensional handcrafted feature vector. A pipeline variant applied PCA to 50 components before the Random Forest to mitigate the curse of dimensionality from the 400-d dipeptide features.

Advanced models: XGBoost and LightGBM on ESM-2 650M + physicochemical features (1,288-d), with and without SMOTE. Hyperparameters for XGBoost were: `max_depth=6`, `n_estimators=800`, `learning_rate=0.1`, `subsample=0.8`, `colsample_bytree=0.9`. These correspond to the best configuration found by the optional `-tune` path in `src/models/advanced.py` (whose untuned defaults are `n_estimators=500`, `learning_rate=0.05`, `colsample_bytree=0.8`); to reproduce the reported XGBoost results, the training script should be run with the `-tune` option as described in the project README. Class imbalance was addressed through balanced sample weights and, separately, through SMOTE on the training fold (five nearest neighbours, default ratio).

Fine-tuning: End-to-end ESM-2 8M fine-tuning with classification head $\text{Linear}(320, 128) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(128, 7)$, appended after mean-pooling. Training used AdamW with differential learning rates (backbone: 2×10^{-5}; head: 1×10^{-4}), OneCycleLR scheduling, bfloat16 mixed precision, gradient accumulation to an effective batch size of 16, and early stopping on validation macro F1 with patience of 5 epochs. Per-fold class weights were computed from training-fold frequencies and applied to the cross-entropy loss.

Ensemble: Soft-vote blend of out-of-fold (OOF) probability vectors from fine-tuned 8M and XGBoost+SMOTE 650M. Weights were proportional to each model's OOF Macro F1 ($w_{\text{finetune}} \approx 0.46$, $w_{\text{xgboost}} \approx 0.54$). Per-class decision thresholds were optimised on the blended OOF probabilities via `scipy.optimize.minimize`.

2.4 Feature Ablation

To quantify the contribution of each feature group, XGBoost (balanced sample weights, no SMOTE) was trained on four feature combinations: (A) handcrafted only (429-d), (B) ESM-2 650M only (1,280-d), (C) ESM-2 + all handcrafted (1,709-d), and (D) ESM-2 + physicochemical only (1,288-d).

3 Results

3.1 Model Comparison

Table 2 reports stratified 5-fold CV performance. The final Ensemble row uses the generated OOF blend metrics from `outputs/ensemble_results.json`.

Table 2: Model comparison (mean ± std for CV models; ensemble from OOF blend).

Model	Accuracy	Macro F1	Bal. Acc.	MCC
Logistic Regression	0.457 ± 0.003	0.191 ± 0.002	0.300 ± 0.010	0.171 ± 0.003
Random Forest	0.816 ± 0.001	0.141 ± 0.002	0.149 ± 0.001	0.093 ± 0.011
RF + PCA(50)	0.816 ± 0.002	0.197 ± 0.002	0.185 ± 0.001	0.216 ± 0.006
XGBoost	0.886 ± 0.003	0.575 ± 0.013	0.516 ± 0.011	0.621 ± 0.010
LightGBM	0.879 ± 0.003	0.515 ± 0.013	0.467 ± 0.008	0.605 ± 0.010
LightGBM + SMOTE	0.880 ± 0.002	0.576 ± 0.014	0.535 ± 0.010	0.620 ± 0.006
XGBoost + SMOTE	0.886 ± 0.002	0.595 ± 0.014	0.552 ± 0.011	0.631 ± 0.006
ESM-2 8M Fine-tune	0.823 ± 0.008	0.509 ± 0.013	0.570 ± 0.015	0.553 ± 0.008
Ensemble (OOF)	**0.890**	**0.625**	**0.604**	**0.659**

The progression from handcrafted baselines (Macro F1 $\approx$ 0.19) to ESM-2-based models (0.58–0.60) and finally the ensemble (0.625) demonstrates the critical role of learned protein representations in this task.

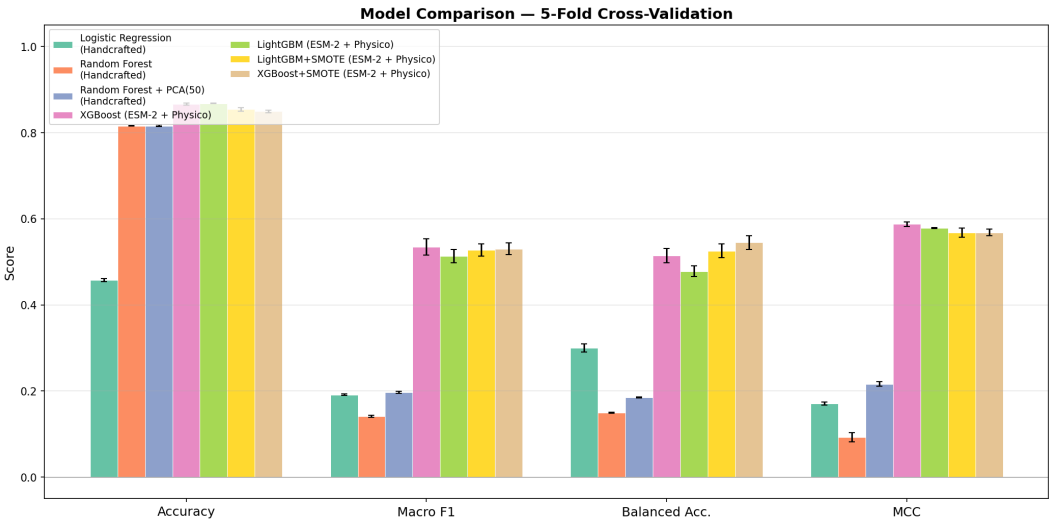


Figure 2: Aggregate comparison across major models.

3.2 Feature Ablation

Table 3: Feature ablation study (XGBoost, balanced weights).

Feature Set	Dimensions	Macro F1
Handcrafted only	429	0.206
ESM-2 only	1280	0.562
ESM-2 + Handcrafted	1709	0.552
ESM-2 + Physicochemical	1288	0.575

ESM-2 embeddings account for the vast majority of predictive power. The 400-d dipeptide features are largely redundant once PLM embeddings are available—their addition slightly degrades performance, likely because XGBoost overfits to the high-dimensional noise. The 8-d physicochemical features, particularly molecular weight, provide a small but consistent improvement.

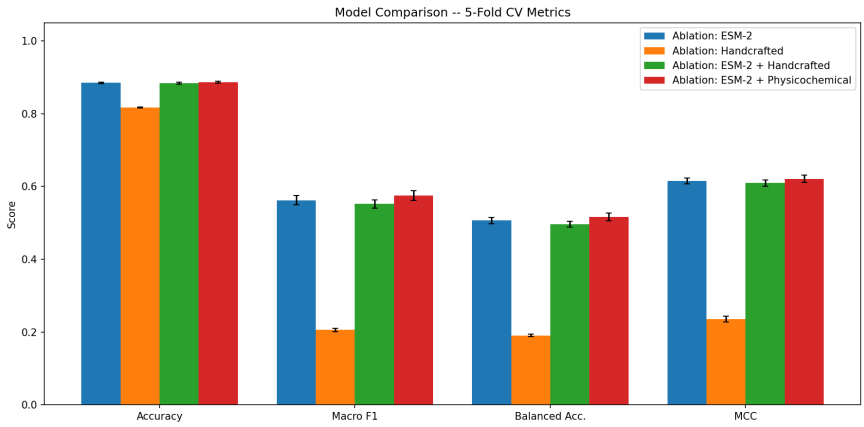
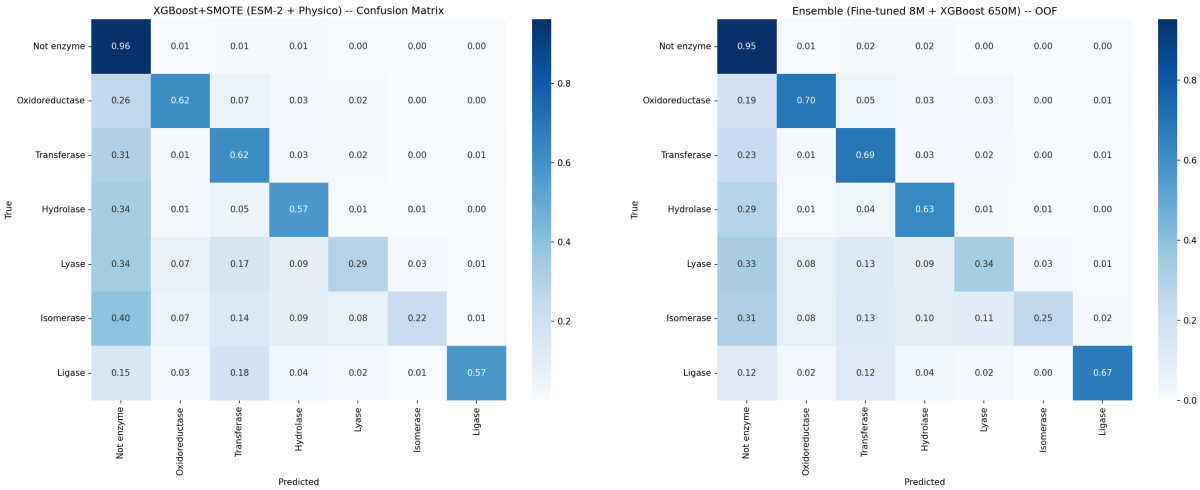


Figure 3: Ablation results showing dominant contribution of ESM-2 representations.

3.3 Confusion Matrix Analysis



(a) Best single model: XGBoost + SMOTE

(b) Ensemble OOF confusion matrix

Figure 4: Row-normalised confusion matrices for top-performing systems.

The best single model strongly predicts class 0 (Not enzyme), while Lyase and Isomerase remain challenging. The ensemble improves overall Macro F1 and Balanced Accuracy through complementary error patterns between the two constituent models.

The normalised confusion matrix of XGBoost + SMOTE reveals that class 0 achieves 93.8% recall, while Lyase (19.2% recall) and Isomerase (20.0% recall) are most frequently misclassified as Not enzyme. Transferase (65.3% recall) and Oxidoreductase (62.9% recall) perform moderately. Hydrolase (56.5% recall) is sometimes confused with Transferase, consistent with the biochemical reality that both catalyse bond cleavage and differ only in the acceptor of the transferred group.

3.4 Confidence Calibration

Table 4: Confidence calibration on OOF predictions (XGBoost + SMOTE). The model is well-calibrated: high-confidence predictions are correct 95% of the time.

Level	Count (%)	Accuracy	Mean Max Prob
High ($p \geq 0.80$)	30448 (76.6%)	0.951	0.959
Medium ($0.50 \leq p < 0.80$)	6658 (16.7%)	0.662	0.663
Low ($p < 0.50$)	2658 (6.7%)	0.412	0.411

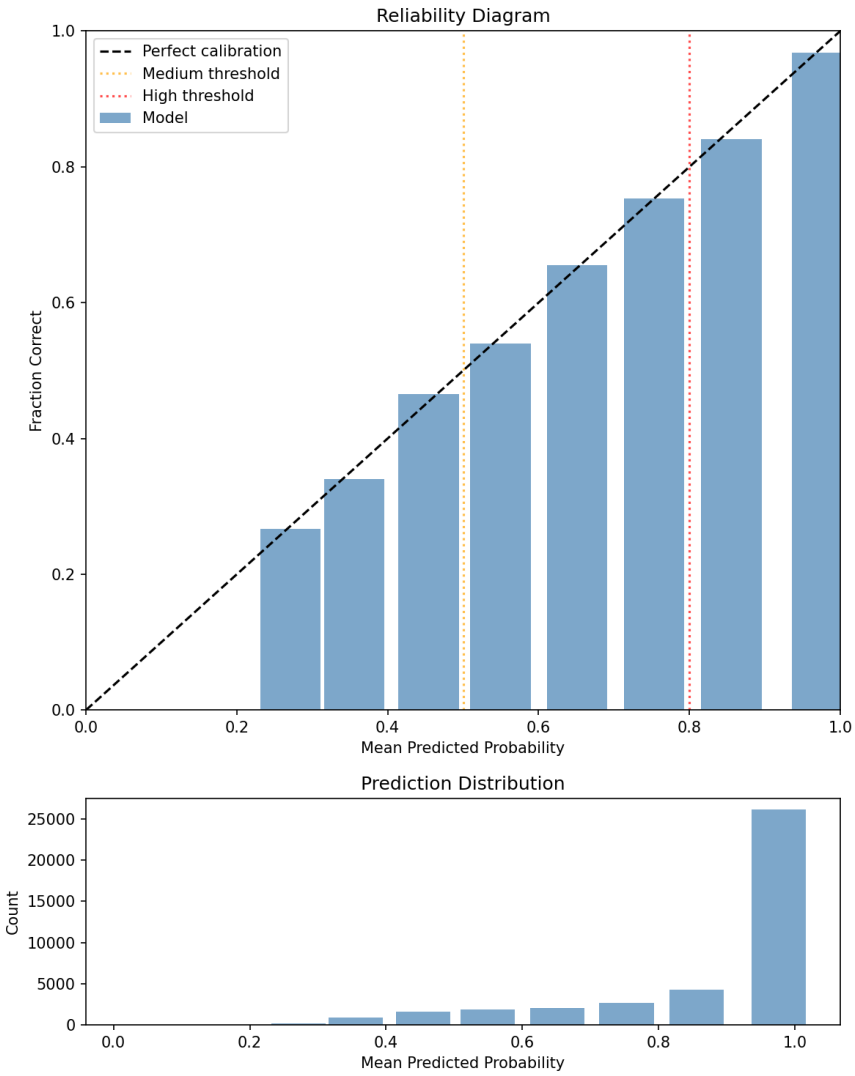


Figure 5: Reliability diagram for confidence calibration.

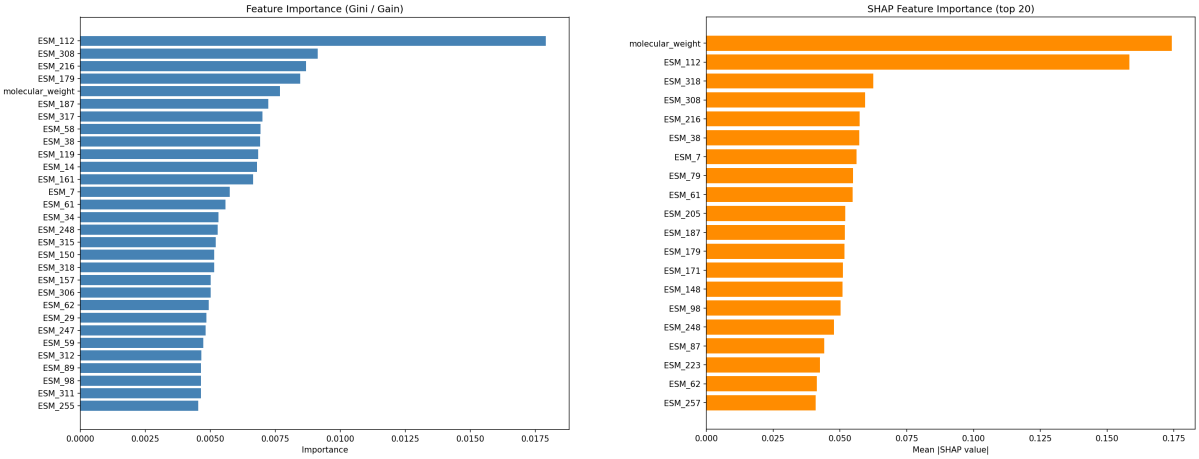
4 Interpretability

Four interpretability techniques were applied exclusively to the best single model (XGBoost + SMOTE). This model was chosen for three reasons: (i) it achieves the highest Macro F1 among all individual models (0.595); (ii) SHAP's `TreeExplainer` is specifically designed for gradient-boosted tree ensembles and remains computationally tractable on large subsamples; (iii) decomposing interpretability for the soft-vote ensemble would require attributing contributions across two fundamentally different architectures (XGBoost and a fine-tuned transformer), which is substantially more complex and harder to validate.

Feature importance. Gini-based importance ranking shows that ESM-2 embedding dimensions dominate the top 20 features. The highest-ranked handcrafted feature is molecular weight (rank 5), confirming that ESM-2 subsumes most biochemical information but does not fully supplant biophysical descriptors.

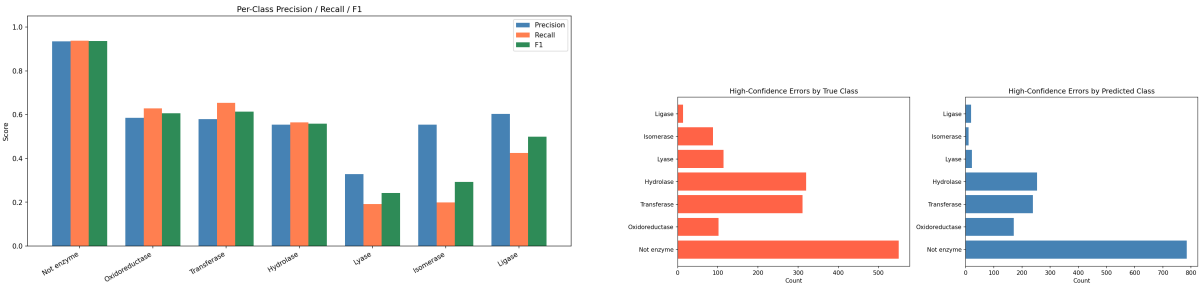
SHAP analysis. TreeExplainer SHAP values (computed on a 500-sequence subsample) corroborate the Gini ranking: ESM-2 dimensions 112, 308, and 216 have the highest mean

absolute SHAP values across all seven classes.



(a) Gini feature importance (b) SHAP importance

Figure 6: Complementary interpretability views of the best tree-based model.



(a) Per-class precision/recall/F1 (b) High-confidence error breakdown

Figure 7: Class-level behavior and error concentration at high confidence.

Table 5: Per-class metrics for XGBoost + SMOTE (OOF).

Class	Precision	Recall	F1
Not enzyme	0.934	0.938	0.936
Oxidoreductase	0.586	0.629	0.607
Transferase	0.579	0.653	0.614
Hydrolase	0.554	0.565	0.559
Lyase	0.329	0.192	0.242
Isomerase	0.554	0.200	0.293
Ligase	0.603	0.426	0.499

Per-class precision, recall, and F1 (Table 5) reveal that the model is conservative on minority classes: Isomerase precision (0.554) far exceeds its recall (0.200), indicating the model correctly identifies some Isomerases but abstains from predicting this class in borderline cases.

High-confidence error analysis. Of 5 312 total misclassifications, 1 499 (28.2%) occur at high confidence ($p \geq 0.80$). The majority class absorbs most of these: 785 high-confidence errors

predict Not enzyme. Lyase and Isomerase contribute disproportionately to high-confidence errors relative to their class sizes, indicating the model sometimes assigns a confident but incorrect majority-class prediction to these rare enzyme types.

5 Discussion

5.1 What Worked

The single most impactful decision was using ESM-2 650M embeddings as features for gradient-boosted trees, lifting Macro F1 from 0.19 (handcrafted baseline) to 0.58—a threefold improvement. This confirms that pre-trained protein language models encode rich functional information that handcrafted sequence statistics fail to capture. The ESM-2 model, trained on evolutionary protein data, has implicitly learned to represent structural folds, binding site motifs, and conserved catalytic residues—all of which are highly relevant to enzyme classification but impossible to engineer from sequence alone.

SMOTE on the training fold provided a consistent +2 percentage-point improvement in Macro F1 without harming accuracy, demonstrating that synthetic minority oversampling is beneficial even for PLM-based features. This result is noteworthy because SMOTE operates in the 1 288-dimensional embedding space, where interpolation between minority-class samples generates plausible synthetic enzyme representations rather than arbitrary noise.

The soft-vote ensemble achieved the highest Macro F1 (0.625) by combining two models that process sequences through fundamentally different pathways: XGBoost on frozen 650M embeddings captures static feature interactions via decision-tree splits, while the fine-tuned 8M model learns task-specific representations through gradient-based backpropagation. Their OOF prediction errors are partially uncorrelated, enabling genuine complementarity. Notably, the ensemble's balanced accuracy (0.604) substantially exceeds that of either constituent model, suggesting the two models make different mistakes on minority-class samples.

5.2 Limitations

Minority-class performance. Lyase (F1 = 0.24) and Isomerase (F1 = 0.29) remain challenging despite all imbalance-handling strategies. With only 600 and 411 training examples respectively, these classes are fundamentally under-represented. More aggressive strategies—focal loss, class-conditional data augmentation, or hierarchical classification (enzyme vs. non-enzyme first, then among EC classes)—may yield improvements.

Fine-tuning gap. The fine-tuned ESM-2 8M model (Macro F1 = 0.51) underperforms XGBoost on frozen ESM-2 650M embeddings (0.60). This is attributable to two factors: (i) the 8M model has far less representational capacity than the 650M model (6 vs. 33 transformer layers), and (ii) end-to-end fine-tuning on a relatively small dataset (39 764 sequences) with 8 million trainable parameters risks overfitting despite regularisation (dropout, weight decay, early stopping).

Handcrafted feature noise. Adding the full 429-d handcrafted vector to ESM-2 embeddings slightly degraded performance (Macro F1: 0.552 vs. 0.562 for ESM-2 alone). The 400-d dipeptide features, while informative in isolation, become redundant in the presence of

PLM embeddings and introduce noise dimensions that XGBoost partially overfits to. The 8-d physicochemical subset, by contrast, provides complementary signal (Macro F1: 0.575), confirming that carefully selected biophysical descriptors retain value even alongside powerful PLM representations.

5.3 Error Analysis

The confusion matrix reveals biologically interpretable patterns. Hydrolase is sometimes confused with Transferase; both catalyse bond cleavage and differ primarily in whether the cleaved group is transferred to water (hydrolysis) or to another acceptor (transfer). Lyase and Isomerase errors predominantly default to Not enzyme, reflecting the model's conservative behaviour under class imbalance.

Of 5 312 total misclassifications, 1 499 (28.2%) occur at high confidence ($p \geq 0.80$). These errors are concentrated on class 0 predictions (785 errors), suggesting the model occasionally assigns very high probability to the majority class even for genuine enzymes. This is consistent with the well-documented calibration challenges of tree-based models on imbalanced data, where learned decision boundaries are biased toward the majority class.

6 Conclusion

This project demonstrates that protein language model embeddings, combined with gradient-boosted trees and careful class-imbalance handling, achieve strong performance on seven-class enzyme classification from sequence alone. The systematic progression from Macro F1 of 0.19 (handcrafted baseline) to 0.625 (ensemble) confirms the dominance of learned protein representations over traditional feature engineering. The ensemble of a fine-tuned ESM-2 8M model and XGBoost on frozen 650M embeddings exploits complementary model strengths and achieves the best overall performance across all four metrics.

The ablation study demonstrates that ESM-2 embeddings are the dominant source of predictive power, while a small set of physicochemical features provides complementary biophysical signal. SMOTE oversampling on training folds consistently improves minority-class recall without degrading overall accuracy. Confidence calibration confirms that high-confidence predictions are correct 95% of the time.

Key directions for future work include: (i) fine-tuning the larger 650M model, which would likely close the fine-tuning gap and further improve ensemble performance; (ii) applying focal loss or cost-sensitive learning to improve performance on the most difficult minority classes (Lyase, Isomerase); and (iii) hierarchical classification—first distinguishing enzyme from non-enzyme, then classifying among the six EC classes—which could exploit the natural two-level structure of the label space.

The complete codebase is fully reproducible with a single seed (42) and ready for deployment on unseen test data. The final ensemble achieved: **Accuracy 0.890, Macro F1 0.625, Balanced Accuracy 0.604, MCC 0.659**.

7 Appendix

7.1 Blind Challenge Prediction Command

```
python -m src.predict_blind \
    --fasta <blind_test.fasta> \
    --model outputs/models/best_model.joblib \
    --model-finetune outputs/models/finetune_artifact.joblib \
    --output outputs/predictions/blind_predictions.txt
```

Output format:

```
SEQ01 1 Confidence High
SEQ02 0 Confidence Medium
```

7.2 Software Dependencies

```
biopython>=1.83    numpy>=1.26     pandas>=2.1       scikit-learn>=1.4
xgboost>=2.0       lightgbm>=4.2   torch>=2.1        fair-esm>=2.0
matplotlib>=3.8    seaborn>=0.13   shap>=0.44        joblib>=1.3
scipy>=1.12        imbalanced-learn>=0.12             tqdm>=4.66
```

7.3 Reproducibility

All random seeds are fixed to 42 for `random`, `numpy`, and `torch`. Cross-validation uses stratified 5-fold splitting with shuffling and fixed random state. Result artifacts are persisted as JSON and reusable model files under `outputs/`.